Homework Assignment 1:
Intro to Particle Physics

All problems are graded on effort only.

Griffiths: Please complete P1.1, P1.2, P1.3, P2.1, P2.2

Problem 1: In lecture we discussed conservation of baryon number (A), strangeness (S), muon number (L_μ), electron number (L_e), and total charge (Q). For each process below, identify one of the quantities above that is not conserved. Note that there are particle listings in the front matter of Griffiths.

(A) $\pi^+ \ + \ p^+ \ \rightarrow \ K^+ \ + \ \Sigma^0$

(B) $\pi^- \ + \ p^+ \ \rightarrow \ \pi^- + \pi^+$

(C) $\pi^- \ \rightarrow \ \mu^- \ + \ \nu_\mu$

(D) $\mu^- \ \rightarrow \ e^- \ + \ \nu_\mu$

(E) $K^0 \ \rightarrow \ \pi^+ \ + \ \pi^-$

Problem 2:

Which of the processes from Problem 1 is allowed only as a weak process?

Problem 3:

(A) Draw the lowest order Feynman diagram for the process:

$$e^+ + e^- \rightarrow \gamma \rightarrow u + \bar{u}.$$

Experimentally, at center-of-mass energies of around 10 GeV, this process produces two distinct jets of particles, the result of hadronic showers of the final state u and $\bar{u}$. (Jump ahead to Griffith's Fig 8.1 if you missed this in lecture.)

(B) Ignoring hadronization and focussing on just the Feynman diagram, is this a QED process or a QCD process?

(C) Add a fundamental QCD vertex to your diagram from (A) for a process that has **three** jets in the final state. Note that jets originate from final state quarks or gluons. There are two possible places to attach the gluon: draw one diagram for each possibility.

(D) Each QED vertex adds a factor of α_{EM} to the rate of the process, and each QCD vertex adds a factor of α_{s}. There are other factors, but let's ignore them. Assume each diagram for a process contributes to the rate without any interference with other diagrams. With these assumptions, what would you expect the ratio of the rate of process (C) to that of process (A) to be? Hint: the LEP experiment used these processes to measure α_s.